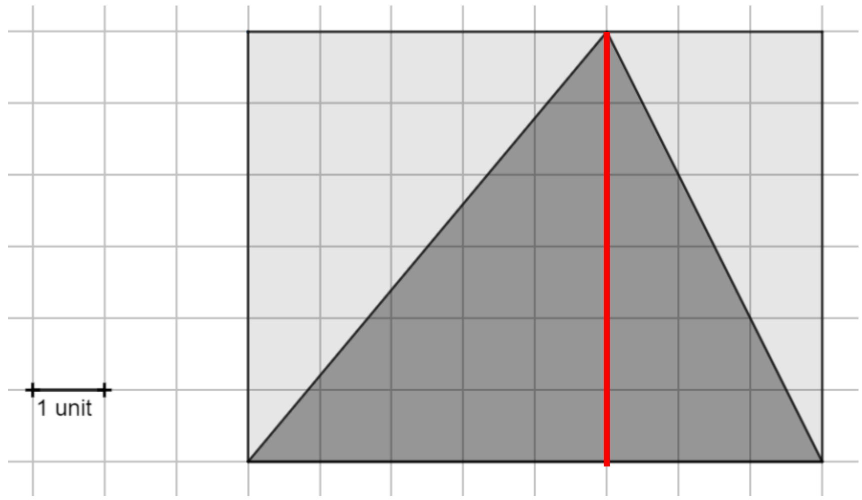
Grade 6 Accelerated Name____________________
Unit 7 Assessment Date_________________
Calculator Period__________

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

1. A rectangle that has a length of 8 units and a width of 6 units with a triangle that is inscribed in the rectangle is shown.



a. Draw a line segment that shows the height of the triangle.

b. Determine the area of the triangle.

Area	24 square units

c. Complete the statements.

The triangle and the rectangle have
- only the same base.
- only the same height.
- both the same base and the same height.

The area of
- all triangles is
- any acute triangle is
- any equilateral triangle is
- any triangle that is inscribed in a rectangle is

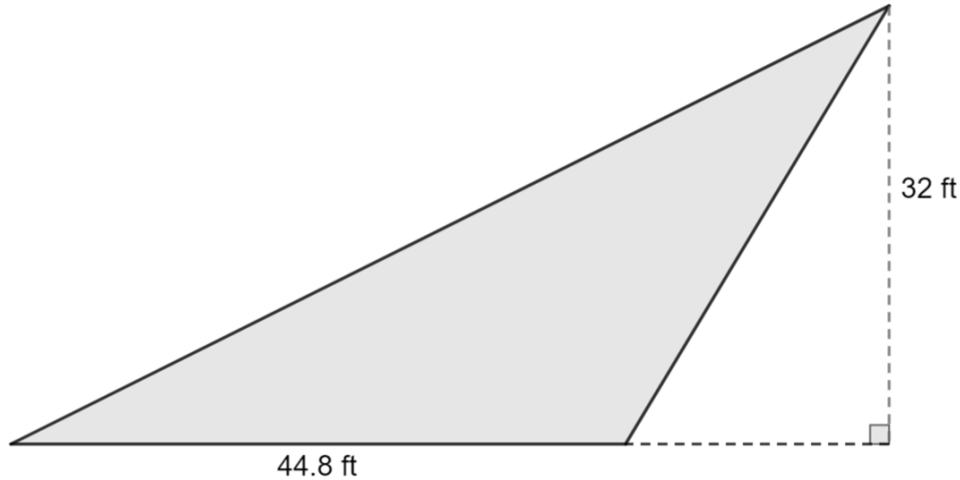
equal to
- bh ,
- $b+h$,
- $\frac{1}{2}bh$,
- $\frac{1}{2}b+\frac{1}{2}h$,

where b represents the base of the triangle and h represents the height of the triangle.

**Students who select, "all triangles is" for the second answer choice should also be marked correct.*

2. A city planner is designing a skateboard park. The park is triangular in shape. The outline of the park is shown with dimensions in feet (ft).

Determine the area, in square feet, of the skateboard park.

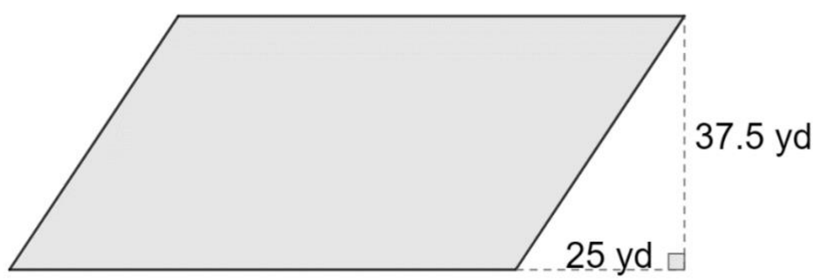


work	
Area	716.8 square feet

3. A quadrilateral with some dimensions, in yards (yd), is shown.

The area of the quadrilateral is 2812.5 square yards.

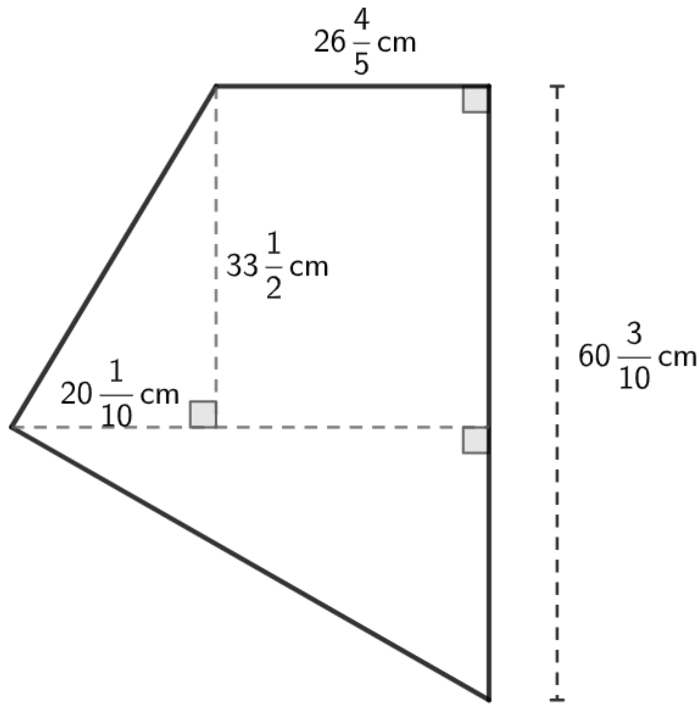
Determine the quadrilateral's missing dimension.



work	
Missing Dimension	75 yards

4. A quadrilateral with dimensions in centimeters (cm) is shown.

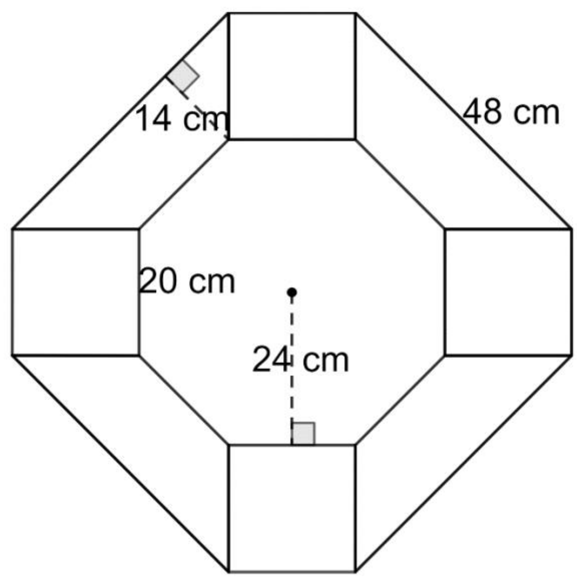
Determine the area, in square centimeters (sq cm), of the quadrilateral.



work

Area	$1862\frac{187}{200}$ or 1862.935 sq cm

5. A sidewalk is created using bricks in the shape of a square, regular octagon, and a trapezoid. The bricks are placed in the pattern with measurements in centimeters (cm) as shown. Determine the area, in square centimeters (sq cm), of the pattern shown.

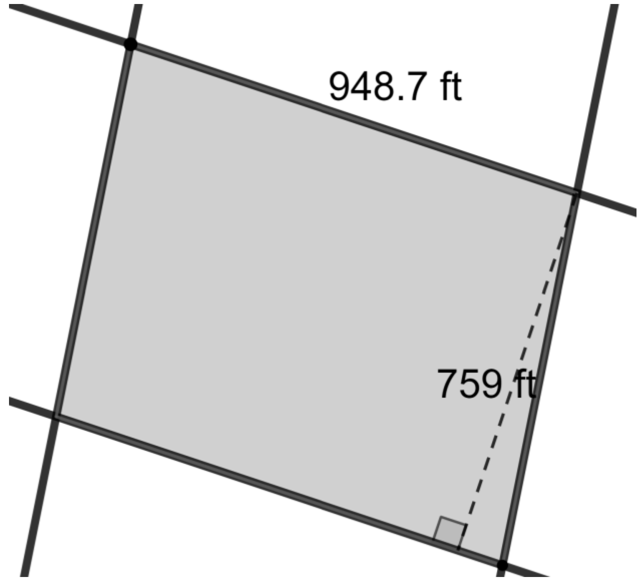


work

Area	5424 sq cm

6. A city planner is redesigning the space between two sets of parallel roads. A diagram of the roads is shown with dimensions given in feet (ft).

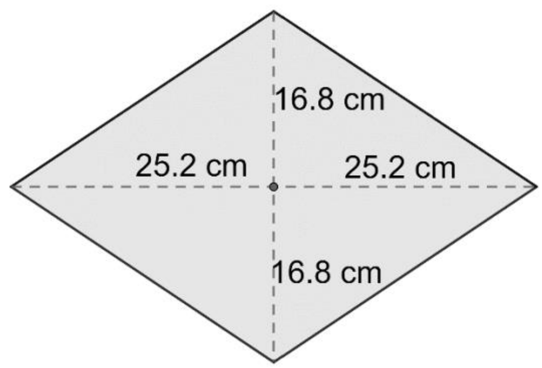
Determine the area between the two sets of parallel roads.



work	
Area	720,063.3 sq ft

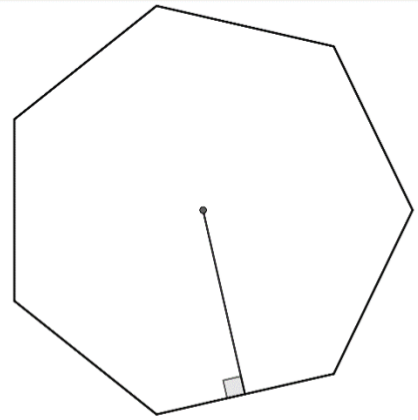
7. A rhombus is shown with measurements in centimeters (cm).

Determine the area, in square centimeters (sq cm), of the rhombus. Show your work.



work	
Area	846.72 sq cm

8. A heptagon is shown, where each side has a measurement of 102 inches (in.). The perpendicular distance from the center to midpoint of a side is 105.91 in.



Determine the area of the heptagon.

work	
Area	37809.87 sq in.